THE COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

DATA COMMUNICATIONS AND PARALLEL PROCESSING **CS 320**

End of term 1 2017/2018 academic year

INSTRUCTIONS

1. Answer ALL questions

QUESTION 1

a) The historic motivation of networking was not to transmit data, explain the historic motivation and reasons behind the motivation (3 marks)

b) Explain the meaning of the following terms as used in data communication:

(i) Frequency (ii) wave length (iii) baud rate (iv) bit rate (4 marks)

c) A data communication system has five (5) major components.

(i) Use a diagram showing these components (3 marks)
(ii) Which component(s) is/are more important than the other(s)? (1 mark)

(iii) Explain the purpose/function of each component (5 marks)

d) What is the major objective of data transmission and how is this objective achieved? (2 marks)

e) Explain with the help of diagrams the difference (s) between a baseband and broadband channel (2 marks)

QUESTION 2

a) Unguided transmission uses antennas that can be directional or directional, explain with the help of simply diagrams difference(s) between directional and directional antennas (4 marks)

b) Microwave transmission is line of sight transmission. Explain what this means (2 marks)

c) A micro wave is travelling at the speed of light for a distance of 5 km. Calculate signal loss due to attenuation (3 marks)

d) A satellite radio wave signal is travelling from a ground station to Geo station and back at the speed of light. Calculate the time it takes for the round trip (3 marks)

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e) Explain purpose(s)/function(s) of the following components of a satellite:

(i) Communication system

(ii) Power system

(iii) Batteries

(iv) Thermal system

(8 marks)

QUESTION 3

a) There are two major reasons why serial transmission is preferred to parallel transmission despite the fact that it is slow than parallel transmission:

(i) Explain the two reasons (4 marks)

(ii) Ethernet technology specifies that data be sent byte big- endian and bit little-endian. Use a four byte word to explain this concept (4 marks)

(iii) Give any four EIA standards for RS -232 C (4 marks)

b) Typical transmission in asynchronous is made up of the following overhead bits:

- 1 start bit
- 1 stop bit
- 1 parity bit

And synchronous transmission is made up of the following overhead bytes: 2 SYN bytes and 2 CRC bytes.

(i) At first sight it seems synchronous transmission has more overheads bits than asynchronous transmission, explain with the help of diagrams why this is not true (4 marks)

(ii)Use a 10 K byte data transmission to compare the efficiencies of the two transmissions (4 marks)

QUESTION 4

a) A noiseless channel has a bandwidth of 3000 Hz (3 kHz) and a signal with two levels. Calculate the maximum bit rate/channel capacity (2 marks)

b) A communications engineer needs to transmit data at a transfer rate of 265k bps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels will he need? (3 marks)

c) Consider a telephone line with a bandwidth of 3 kHz and a signal to noise ratio which is almost zero.

(i) Calculate the Channel transfer rate (3 marks)

(ii) Explain your answer (2 marks)

d) The telephone system has a signal to noise ratio of approximately 30 dB and an analog bandwidth of 3000Hz (3 kHz). Convert the decibels to units (2 marks)

e) During an AM, the maximum amplitude reached was 75 V and the minimum reached was 10 V. Calculate the Modulation depth or index (2 marks)

f) During long distance transmission, a communication system was using ASK and the transmitted bits were 1011001010. Use a diagram to show how ASK operates; clearly show the transmitted information (modulating signal), the carrier frequency and the modulated signal. Use carrier frequency of your own choice . (5 marks)

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